

Ratio, Root, and Alternating Series Tests



The Ratio Test

- 1 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
 - 2 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ converges.
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The Root Test

- 3 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.
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Alternating series and the Alternating Series Test

- 4 State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.
 - 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.
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Alternating series error bound

- 6 Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.
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Absolute vs. conditional convergence

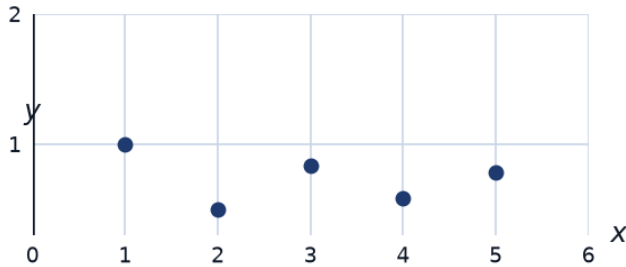
- 7 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.
 - 8 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges absolutely, conditionally, or diverges.
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More Ratio Test practice

- 9 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.
 - 10 Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.
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Partial sums of an alternating series (visualizing oscillating convergence)

- 11 Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.



Determining the number of terms needed for a given accuracy

- 12 For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

Mixed practice: classify each series

- 13 Classify each series as convergent (state absolutely or conditionally) or divergent:

- a $\sum \frac{(-1)^n}{\sqrt{n}}$
- b $\sum \frac{(-1)^n n^2}{n^3 + 1}$
- c $\sum \frac{(-1)^n}{n!}$

The Root Test with more complex expressions

- 14 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$ converges.
- 15 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$ converges.

Why the Ratio Test fails for p-series

- 16 Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p .

Combining multiple tests in one problem

- 17 Determine whether $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$ converges absolutely, conditionally, or diverges.
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Multiple choice — easy 15

- 18 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.
- a. 3
 - b. $\frac{1}{3}$
 - c. $\frac{4}{3}$
 - d. $\frac{1}{4}$
- 19 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{8^n}$.
- a. 8
 - b. $\frac{9}{8}$
 - c. $\frac{1}{8}$
 - d. $\frac{1}{9}$
- 20 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{14^n}$.
- a. 14
 - b. $\frac{1}{14}$
 - c. $\frac{1}{15}$
 - d. $\frac{15}{14}$
- 21 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{15^n}$.
- a. 15
 - b. $\frac{1}{16}$
 - c. $\frac{16}{15}$
 - d. $\frac{1}{15}$

22 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{10^n}$.

a. 10

b. $\frac{1}{11}$

c. $\frac{1}{10}$

d. $\frac{11}{10}$

23 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{19^n}$.

a. $\frac{1}{20}$

b. $\frac{20}{19}$

c. $\frac{1}{19}$

d. 19

24 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{13^n}$.

a. $\frac{1}{14}$

b. 13

c. $\frac{14}{13}$

d. $\frac{1}{13}$

25 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{5^n}$.

a. $\frac{6}{5}$

b. 5

c. $\frac{1}{6}$

d. $\frac{1}{5}$

26 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{11^n}$.

a. $\frac{1}{11}$

b. $\frac{1}{12}$

c. 11

d. $\frac{12}{11}$

27 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{20^n}$.

a. $\frac{1}{21}$

b. 20

c. $\frac{21}{20}$

d. $\frac{1}{20}$

28 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{4^n}$.

a. $\frac{5}{4}$

b. 4

c. $\frac{1}{5}$

d. $\frac{1}{4}$

29 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{12^n}$.

a. $\frac{13}{12}$

b. $\frac{1}{13}$

c. $\frac{1}{12}$

d. 12

30 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{17^n}$.

a. $\frac{18}{17}$

b. $\frac{1}{18}$

c. $\frac{1}{17}$

d. 17

31 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{16^n}$.

a. $\frac{1}{16}$

b. $\frac{1}{17}$

c. 16

d. $\frac{17}{16}$

32 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{6^n}$.

a. $\frac{1}{7}$

b. $\frac{1}{6}$

c. $\frac{7}{6}$

d. 6

Multiple choice — medium 15

33 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{4n+4}{11n-7} \right)^n \right|}$.

a. $\frac{4}{11}$

b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

34 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{5n+8}{10n+3} \right)^n \right|}$.

a. 1

b. 2

c. $\frac{1}{2}$

d. $\frac{3}{2}$

35 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n-7}{12n+4} \right)^n \right|}$.

a. $\frac{12}{7}$

b. $\frac{7}{12}$

c. $\frac{19}{12}$

d. $\frac{13}{12}$

36 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{3n+4}{8n-6} \right)^n \right|}$.

a. $\frac{7}{8}$

b. $\frac{8}{3}$

c. $\frac{3}{8}$

d. $\frac{11}{8}$

37 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{5n+7}{11n+4} \right)^n \right|}$.

a. $\frac{5}{11}$

b. $\frac{11}{5}$

c. $\frac{21}{22}$

d. $\frac{16}{11}$

38 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{4n+6}{9n-2} \right)^n \right|}$.

a. $\frac{9}{4}$

b. $\frac{4}{9}$

c. $\frac{17}{18}$

d. $\frac{13}{9}$

39 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{2n-1}{11n-2} \right)^n \right|}$.

a. $\frac{2}{11}$

b. $\frac{15}{22}$

c. $\frac{11}{2}$

d. $\frac{13}{11}$

40 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{9n+4}{12n+3} \right)^n \right|}$.

a. $\frac{3}{4}$

b. $\frac{7}{4}$

c. $\frac{4}{3}$

d. $\frac{5}{4}$

41 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{n+3}{7n+8} \right)^n \right|}$.

a. $\frac{9}{14}$

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{7}$

42 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{6n-6}{12n-3} \right)^n \right|}$.

a. $\frac{3}{2}$

b. $\frac{1}{2}$

c. 2

d. 1

43 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n-7}{16n+2} \right)^n \right|}$.

a. $\frac{7}{16}$

b. $\frac{16}{7}$

c. $\frac{15}{16}$

d. $\frac{23}{16}$

44 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{8n-6}{13n+7} \right)^n \right|}$.

a. $\frac{21}{13}$

b. $\frac{8}{13}$

c. $\frac{29}{26}$

d. $\frac{13}{8}$

45 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{6n-6}{8n-2} \right)^n \right|}$.

a. $\frac{3}{4}$

b. $\frac{4}{3}$

c. $\frac{7}{4}$

d. $\frac{5}{4}$

46 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n+6}{11n-9} \right)^n \right|}$.

a. $\frac{25}{22}$

b. $\frac{11}{7}$

c. $\frac{18}{11}$

d. $\frac{7}{11}$

47 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{8n-5}{12n-4} \right)^n \right|}$.

a. $\frac{3}{2}$

b. $\frac{2}{3}$

c. $\frac{5}{3}$

d. $\frac{7}{6}$

Multiple choice — hard 15

48 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 5}{n^1}$ converges absolutely, conditionally, or diverges.

a. Diverges

b. Cannot be determined without more information

c. Conditionally convergent

d. Absolutely convergent

49 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+6}$ converges absolutely, conditionally, or diverges.

a. Cannot be determined without more information

b. Diverges

c. Conditionally convergent

d. Absolutely convergent

- 50 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 3}{n^3}$ converges absolutely, conditionally, or diverges.
- a. Conditionally convergent
 - b. Cannot be determined without more information
 - c. Diverges
 - d. Absolutely convergent
- 51 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 4}{n^1}$ converges absolutely, conditionally, or diverges.
- a. Absolutely convergent
 - b. Cannot be determined without more information
 - c. Diverges
 - d. Conditionally convergent
- 52 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 7}{n^{\frac{3}{2}}}$ converges absolutely, conditionally, or diverges.
- a. Diverges
 - b. Cannot be determined without more information
 - c. Conditionally convergent
 - d. Absolutely convergent
- 53 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{8^n}{n!}$ converges.
- a. Diverges ($L = \infty$)
 - b. Converges by the Alternating Series Test
 - c. Inconclusive ($L = 1$)
 - d. Converges ($L = 0$)

54 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

- a. Inconclusive ($L = 1$)
- b. Converges by the Alternating Series Test
- c. Diverges ($L = \infty$)
- d. Converges ($L = 0$)

55 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{6^n}$ converges.

- a. Converges ($L = 0$)
- b. Converges by the Alternating Series Test
- c. Diverges ($L = \infty$)
- d. Inconclusive ($L = 1$)

56 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{2^n}$ converges.

- a. Diverges ($L = \infty$)
- b. Converges by the Alternating Series Test
- c. Inconclusive ($L = 1$)
- d. Converges ($L = 0$)

57 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{7^n}{n!}$ converges.

- a. Inconclusive ($L = 1$)
- b. Converges ($L = 0$)
- c. Diverges ($L = \infty$)
- d. Converges by the Alternating Series Test

58 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n}{n+3}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+3}\right)^n$.

- a. 3
- b. 9
- c. 8
- d. 10

59 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{3n}{n+5}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{3n}{n+5}\right)^n$.

- a. 4
- b. 2
- c. 5
- d. 3

60 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{5n}{n+3}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{5n}{n+3}\right)^n$.

- a. 4
- b. 5
- c. 3
- d. 6

61 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+1}\right)^n$.

- a. 1
- b. 8
- c. 10
- d. 9

62 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{5n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{5n}{n+1}\right)^n$.

a. 5

b. 6

c. 4

d. 1